From statistical perspective → a probability distribution is theoretical model that describes how a random variable varies

Population → entire collection of objects under consideration

while population parameters are unknown, we calculate similar quantities in sample.
So statistical interference → estimating population parameters and assessing precision of those estimates using sample statistics

t- distribution → looks similar to normal distribution, but tails tend to be a little thicker

→ the extra variability is controlled by an integer number called degrees of freedom

↓

smaller this number, more spread out t-distribution

sample mean m_Y → is a good point estimate of the population mean $E(Y)$
It is helpful to know how reliable this estimate is, that is how much sampling uncertainty is associated with it.

Degrees of freedom

↳ Roughly → each data observation provides one degree of freedom but we lose a degree of freedom for each population parameter that we have to estimate

$$t\text{-statistic} = \frac{m_Y - E(Y)}{S_Y / \sqrt{n}}$$

m_Y → sample mean
$E(Y)$ → value of population mean in null hypothesis
S_Y → sample standard deviation
n → sample size

The significance level → dictates the critical value(s) for the test beyond which an observed t-statistic leads to rejection of null hypothesis in favour of the alternative

Short Notes RTSM

$$\hat{\beta}_1 = \frac{\Sigma y_i(x_i - \bar{x})}{\Sigma(x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}} \qquad \bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$$

$$\underset{\sim}{Y} = X\underset{\sim}{B} + \underset{\sim}{\varepsilon} \qquad \hat{\underset{\sim}{y}} = X\hat{\underset{\sim}{B}}$$

Subspace

↳ check whether $0 \in S$ & $x + ay \in S$ $\forall x, y \in S$ & $a \in R$

Span → $sp\{v_1, v_2 \dots v_k\} = \sum_{i=1}^{k} c_i v_i$ $\quad c_i \in R$ → A span is always a subspace

Basis → span of linearly independent vectors; dimension → no. of linearly independent vectors in basis

orthogonal vectors → $u, v \in V$ $\quad u^T v = 0$

orthogonal complement → $S \subseteq V$ then orthogonal complement of S, is $S^\perp$

$$S^\perp = \{v \mid v \in V, u^T v = 0, \forall u \in S\}$$

$S^\perp$ ka har vector S ke har vector se orthogonal hoga

$$\dim(S^\perp) = \dim(V) - \dim(S)$$

Some points ⇒ ① Basis is not unique ② Elements of basis need not be orthogonal to each other
③ linear independence need not imply orthogonality
④ Orthogonality implies independence

Projection Matrix → square matrix, idempotent, symmetric

Rank → no. of non-zero eigenvalues.
$Tr(A)$ → sum of eigenvalues

① If $S \subseteq V$ then projection matrix of subspace S is P_S
a) $P_S v = v$ if $v \in S$
b) $P_S v \in S$ for all $v \in V$

Orthogonal projection matrix → P_S is orthogonal projection matrix of $S \subseteq V$ if $(I - P_S)$ is projection matrix of $S^\perp \subseteq V$ too

→ If $\{v_1, v_2 \dots v_k\}$ is an orthonormal basis of subspace $S \subseteq V$ then orthogonal projection matrix of S is $P_S = \Sigma v_i v_i^T$

→ In regression $\hat{y}$ eventually becomes orthogonal projection of $y \in R^m$ in subspace $S = sp\{1, x\}$

Column space

$$A = [a_1\, a_2 \dots a_n] \qquad C(A) = sp\{a_1, a_2 \dots a_n\}$$

Hence row space $R(A) = C(A^T)$

Properties : ① $C(A:B) = C(A) + C(B)$
② $C(AB) \subseteq C(A)$
③ $\dim[C(A)] = \text{Rank}(A)$
④ $C(AA^T) = C(A) \Rightarrow \text{Rank}(AA^T) = \text{Rank}(A)$

Quadratic forms

x^TAx $A \to$ square symmetric matrix

① pd if $x^TAx > 0$ for $x \neq 0 \in R^n$
② psd if $x^TAx \geq 0$ " " $\in R^n$

Properties a) If A is pd then $|A| > 0$ b) If A is psd then $|A| \geq 0$

Generalised Inverse

$AGA = A$

Properties
① If A is $m \times n$ then A^- is $n \times m$
② A^{-1} is not unique

Imp
③ For A, projection matrix of $C(A)$ is AA^{-1}
④ For A, orthogonal projection matrix of $C(A)$ is $A(A^TA)^{-1}A^T$ $\to$ stat

Dispersion Matrix (Variance Covariance Matrix)

$var(X) = E(X^2) - [E(X)]^2 = E(X - E(X))^2$

$cov(X, Y) = E(X - E(X))(Y - E(Y)) \to E(X - \mu_x)(Y - \mu_y)$

Note ① $cov(U_p, U_q) = cov(U_i, V_j)_{p \times q}$
② $cov(X+b, Y+c) = cov(X, Y)$

$D(\underset{\sim}{X}) = E[\underset{\sim}{X} - E(\underset{\sim}{X})](\underset{\sim}{X} - E(\underset{\sim}{X}))^T$
$= E[(\underset{\sim}{X} - \underset{\sim}{\mu})(\underset{\sim}{X} - \underset{\sim}{\mu})^T]$
$= E(\underset{\sim}{X}\underset{\sim}{X}^T) - \underset{\sim}{\mu}\underset{\sim}{\mu}^T$

Some Important Results
① $E(l^TX) = l^T\mu$
② $E(A\underset{\sim}{X}) = A\underset{\sim}{\mu}$
③ $D(l^TX) = l^T\Sigma l$
④ $D(AX) = A\Sigma A^T$ and $\boxed{cov(AX, BX) = A\Sigma B^T}$
⑤ $\Sigma = \Sigma^T$

$D(X) \to$ P.S.D matrix & $\Sigma = \Sigma^T$

Theorem 1 : $X \to$ random vector ; $E(X) = \mu$ & $D(X) = \Sigma$

$E(X^TAX) = tr(\Sigma A) + \mu^TA\mu$ $\to$ $\boxed{\text{see proof once}}$

Ex $x_i \overset{iid}{\sim} N(0,1)$ $Q_1 = x^Tx = \Sigma x_i^2$; $E(Q_1) = \sum_{i=1}^{n} x_i^2 = n$
$y_i \overset{iid}{\sim} N(\mu, 1)$ $Q_2 = y^Ty = \Sigma y_i^2$ $E(Q_2) = \underset{\text{degrees of freedom}}{n} + \underset{\text{Non central parameter (Mean shifted)}}{\mu^T\mu}$

Let $A = I_n$
$E(\Sigma x_i^2) = \Sigma E(x_i^2) = \underbrace{\Sigma\sigma_i^2}_{tr(\Sigma I_n)} + \underbrace{\Sigma\mu_i^2}_{\mu^T I_n \mu}$

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<u>Theorem 2</u> $\quad X \rightarrow$ Random Vector ; $E(X) = \mu$, $D(X) = \Sigma$
then
$$P(X-\mu) \in C(\Sigma) = 1$$

In simple words $\rightarrow$ probability of deviation / fluctuation belonging to column space of variance-covariance matrix is 1

Cochran's theorem $\Rightarrow$ If $X \sim N(\mu, I_m)$ then X^TAX has chi-squared distribution iff A is idempotent.
DOF = Rank (A) & NCP = $\mu^T A \mu$

Theorem : Let $X \sim N(\mu, I_m)$; A $\rightarrow$ symmetric $\quad CA = 0$ then X^TAX & CX are independently distributed

Trace (A) = sum of diagonal elements = sum of eigen values

$|A|$ = product of eigen values

If $X \sim N(\mu, I_m)$ then X^TAX is a χ^2 distribution iff A is idempotent. Also $X^TAX \sim \chi^2$
dof = Rank(A)
ncp = $\mu^T A \mu$

Matrix representation

$$\underset{\sim}{Y} = X\beta + \underset{\sim}{\varepsilon} \qquad \varepsilon \sim N(0, \sigma^2 I_m)$$
$$\underset{\sim}{Y} \sim (X\underset{\sim}{\beta}, I_m\sigma^2)$$

$X^T\underset{\sim}{Y} = (X^TX)\underset{\sim}{\beta} \rightarrow$ Normal Equation

$$\underset{\sim}{\hat{\beta}}_{LS} = (X^TX)^{-1}X^TY$$

Maximum likelihood estimation $\rightarrow$ method of estimating parameters under assumption that data we have is most probable

$\hat{\beta}_{MLE} = \hat{\beta}_{LS} \Rightarrow$ assuming that errors are normally distributed

Gauss Markov Model : $Y = X\underset{\sim}{\beta} + \underset{\sim}{\varepsilon}$
$$\varepsilon \sim N(\underset{\sim}{0}, \sigma^2 I_m)$$
$\underset{\sim}{Y} \sim N(X\underset{\sim}{\beta}, \underline{\sigma^2 \underset{\sim}{Y}}) \rightarrow$ holds true for all linear models
$\hookrightarrow$ different, dependent kahi ho skta h y_i

$$\hat{y} = X\hat{\beta} = X[(X^TX)^{-1}X^T\underset{\sim}{Y}]$$
$= P_X\underset{\sim}{Y}$; $P_X = [X(X^TX)^{-1}X^T] \rightarrow$ orthogonal projection matrix of column space of X

Note:- ① For matrix A, orthogonal projection matrix of C(A) is $A(A^TA)^{-1}A^T$
② We can't observe true error, but estimated error $\underset{\sim}{e} = \underset{\sim}{Y} - \underset{\sim}{\hat{Y}} = Y - P_X\underset{\sim}{Y} = \underset{\sim}{Y}[I_m - P_X]$

① $P_x = P_x^T$ (symmetric) ② $P_x = P_x^2$ (Idempotent) ③ $P_x(I_n - P_x) = \underset{\sim}{0}$

④ $\underset{\sim}{e}^T \underset{\sim}{\hat{y}} = 0 \rightarrow$ Error is normal to $\hat{y}$

$$\underset{\sim}{\hat{\beta}} \sim N(\beta, (X^T X^{-1})\sigma^2)$$

Note $\Rightarrow$ Observed error and predicted error are independentally distributed

Simple linear regression

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

If $Z \sim N(\mu, \Sigma)$ then $AZ \sim N(A\mu, A\Sigma A^T)$

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right)$$

$$\hat{\beta}_0 \sim N\left(\beta_0, \sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)\right)$$

Hypothesis testing

Algorithm

① $\tilde{\beta}_1 = \frac{S_{xy}}{S_{xx}} \rightarrow$ from data ; $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$

② $\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right)$

③ $\frac{\hat{\beta}_1 - \beta_1}{\sqrt{\sigma^2 / S_{xx}}} \sim N(0,1) \Rightarrow \frac{\hat{\beta}_1 - b_1}{\sqrt{\sigma^2 / S_{xx}}} \sim N(0,1)$ under H_0

but $\sigma^2 \rightarrow$ unknown

$$\hat{\sigma}^2 = \frac{1}{n-2}\left(S_{yy} - \frac{S_{xy}^2}{S_{xx}}\right)$$

④ $T = \frac{\hat{\beta}_1 - b_1}{\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}} \sim t_{n-2}$ under H_0

$\downarrow$

t-distribution

If $|T_{observed}| > T_{\alpha/2, n-2} \rightarrow$ we reject H_0 in favour of H_1

$\downarrow$

Find Tvalue from t-table with DOF $= n-2$ and significance level $= \alpha/2$

If $|T_{obs}| < T_{\alpha/2, n-2} \rightarrow$ given data we failed to reject H_0 in favour of H_1 at size $\alpha = 0.05$

$$\hat{y} \sim N\left(\beta_0 + \beta_1 x_0,\ \sigma^2\left(\frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}\right)\right)$$

$$\hat{\sigma}^2 = \frac{SS_{Res}}{n-2} = MS_{Res} \rightarrow \text{Model dependent estimate of } \sigma^2$$

$$SS_{Res} = SS_T - \hat{\beta}_1 S_{xy} \qquad SS_T = \sum_{i=1}^{n}(y_i - \bar{y})^2$$

$\hookrightarrow$ get SS_{Res} from here

$$\hat{\sigma}^2 = \frac{\overbrace{S_{yy} - \frac{S_{xy}^2}{S_{xx}}}^{SS_{Res}}}{n-2}$$

$$\lambda = \frac{\beta_1^2 S_{xx}}{\sigma^2} \rightarrow \qquad e^T e \sim \chi^2_{\substack{DOF = \text{Rank of } (I_n - P_x) \\ ncp = 0}}$$

$$\text{Rank}(A-B) = \text{Rank}(A) - \text{Rank}(B)$$

Hypothesis testing

$$\hat{\beta}_1,\ \hat{\beta}_0,\ \hat{\sigma}^2$$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$\beta_1 \quad \beta_0 \quad \sigma^2$$

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right) \qquad \hat{\beta}_0 \sim N\left(\beta_0, \sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)\right)$$

$$\frac{ee^T}{n-2} \sim \chi^2_{\substack{dof = n-2 \\ ncp = 0}}$$

$$\text{Var}(\hat{y}_0) = \text{Var}(\hat{\beta}_0 + \hat{\beta}_1 x_0)$$

$$= \text{Var}(\bar{y} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 x_0)$$

$$= \text{Var}(\bar{y} + \hat{\beta}_1 (x_0 - \bar{x}))$$

$$= \text{Var}\left(\frac{1}{n} \underset{\sim}{1}^T \underset{\sim}{y} + \frac{S_{xy}}{S_{xx}}(x_0 - \bar{x})\right)$$

$$=$$

<u>Bivariate Normal</u>

$$\sim (\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho_{xy})$$

$$Y|X \sim N\left(\mu_y + \rho \frac{\sigma_y}{\sigma_x}(x - \mu_x), \sigma_y^2(1-\rho^2)\right)$$

① $H_0 : \rho = 0$

$H_1 : \rho \neq 0$

$$r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}$$

$$T = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \sim t_{n-2}$$

② $H_0 : \rho = \rho_0 \neq 0$

$H_1 : \rho \neq \rho_0$

$$\left[\begin{array}{l} z = \frac{1}{2}\ln\left(\frac{1+r}{1-r}\right) \\ \mu_0 = \frac{1}{2}\ln\left(\frac{1+\rho_0}{1-\rho_0}\right) \\ \sigma_z^2 = \frac{1}{n-3} \end{array}\right.$$

$$w = \frac{z - \mu_0}{\sigma_z}$$

MSE (Mean square error) $= \frac{\sum (y_i - \hat{y}_i)^2}{n-2} = \frac{SSE}{n-2}$

MSR (Regression mean square) $= \frac{\sum (\hat{y}_i - \bar{y})^2}{1} = SSR$

$SS_{Res} = \sum e_i^2 = \sum (y_i - \hat{y})^2$ $\qquad SS_R = \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2$

$\hookrightarrow \hat{\beta}_1 S_{xy}$

$SS_T = \sum (y_i - \bar{y})^2$

$SS_{Res} = SS_T - \hat{\beta}_1 S_{xy} \Rightarrow SS_T = SS_R + SS_{Res}$

$\hat{\sigma}^2 = \frac{SS_{Res}}{n-2} = MS_{res} \rightarrow$ residual mean square

$$t_0 = \frac{\hat{\beta}_1 - \beta_{10}}{\sqrt{MS_{Res} / S_{xx}}}$$

Reject null hypothesis if $|t_0| > t_{\alpha/2, n-2}$

$se(\hat{\beta}_1) = \sqrt{\frac{MS_{Res}}{S_{xx}}} \rightarrow$ Standard error of slope

For intercept

$$t_0 = \frac{\hat{\beta}_0 - \beta_{00}}{se(\hat{\beta}_0)}$$

$$se(\hat{\beta}_0) = \sqrt{MS_{Res}\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)}$$

Hypothesis testing on slope and intercept

$\hat{\beta}_1 \rightarrow$ normally distributed with mean β_1 and variance σ^2 / S_{xx}

to statistic

$$Z_0 = \frac{\hat{\beta}_1 - \beta_{10}}{\sqrt{\sigma^2 / S_{xx}}} \sim N(0$$

σ^2 → population parameter → rarely know its true value → the best we can do is estimate it.

variance → how spread out

Mean square error

$$MSE = \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{n-2}$$

→ average of squares of errors

→ observed - predicted

MSE → estimator

↳ procedure for estimating an unobserved quantity